

Anisotropy in Faraday instabilities of a shallow conducting fluid

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PACS 47.35.Bb – Gravity waves

PACS 47.15.Fe – Stability of laminar flows

PACS 47.15.K- – Inviscid laminar flows

Abstract – In this work, we study the impact of temporally varying gravity on an idealized shallow conducting fluid with a free surface. We formulate the problem in terms of the Shallow Magnetohydrodynamic Equations (SMHD) which allows the study of the resulting anisotropy in the context of a Faraday instability. We simplify the linear stability problem into a Mathieu equation by assuming gravity is periodic in time. Using the existing theory of Mathieu’s equation, we find that anisotropy occurs, with wave-vectors parallel to the imposed magnetic field stabilized. We describe the regions of instability in wavenumber space.

Introduction. – In the 1830s Faraday reported on simple experiments performed on the dynamics of fluids overlying a vibrating plate, which led to the emergence of complex patterns in the fluid. These patterns depended on the amplitude and frequency [1] of the oscillations.¹ During these experiments Faraday discovered his eponymous instability after observing the surface of the fluid becoming unstable once the forcing frequency exceeded a critical value. The theoretical description of his findings dates back to the 1950s with the work of Benjamin and Ursell [3]. These authors performed a linear analysis that introduced the notation of Parametric Resonance, with a quantitative description via the Mathieu equation. The Mathieu equation is, in itself, a classical problem in applied mathematics with a wide array of available results [4].

The linear theory of Faraday instabilities has been extended in several different physical directions, for example to include viscosity and viscoelasticity [5–8]. The theory has been extended to the weakly nonlinear domain [9], and through perturbation methods [10]. However, the limitations of purely analytical approaches have meant that much of our understanding of Parametric Instability has come from numerical experiments. This has included work in two and three dimensions [11–14], and extending the classical problem to include the Coriolis force [15], surface tension [16], magnetic fields [17], and stochasticity [18]. Much has been learned through physical experiments as

well [19–21].

Paul and Kumar [17] studied the stability of parametrically driven surface waves in the context of liquid metals, subject to a uniform vertical magnetic field parallel to an oscillating gravitational force. They find the magnetic field to be stabilizing. Specifically, they find the amplitude of the oscillating part of gravity must increase to produce a parametric instability as the strength of the background magnetic field increases. Having the magnetic field aligned with the gravitational force implies that no spatial anisotropy becomes apparent.

Our investigation concerns a similar setting, namely a single shallow layer of incompressible, perfectly conducting fluid with a free surface subject to vertically oscillating gravity. The difference lies in the direction of the uniform magnetic field, which we pick to be horizontal, hence perpendicular to the gravitational force. In doing so, we introduce the possibility of anisotropy in the instability, due to relative orientation of the wavenumber and the horizontal magnetic field. In our use of an infinite electrical conductivity, we can apply our problem to astrophysical plasmas since this is not what is typically used for laboratory experiments. In the context of the solar tachocline, the configuration of an thin slab of perfectly electrically conducting fluid in a constant horizontal magnetic field has been formulated and motivated under constant gravity [22]. Various studies of instability were made in this regime to approximate the dynamics of the solar tachocline [23, 24].

In the following sections, we incorporate sinusoidally

¹For the interested reader, Fauve provides details of the experiments from a modern perspective [2].

varying gravity into the shallow magnetohydrodynamic (SMHD) equations with a constant horizontal magnetic field. Linearizing about a still slab of fluid, we reduce the governing equations to a Mathieu Equation, which describes the evolution of the free surface in spectral space. The stability of this equation has been extensively studied in the literature [4], and we modify these approaches for our specific problem. We find that we can map the stability tongues of the Mathieu equation onto bounded regions in the spectral space of the system. We then evolve the Mathieu equation for one set of parameters and several choices of wavenumber to confirm the predicted unstable regions, and to provide more detail on the structure of these regions via Lyapunov exponents.

The Governing Equations and Linearization. –

The shallow magnetohydrodynamic (SMHD) equations describe the evolution of a barotropic layer of inviscid, incompressible and perfectly electrically conducting fluid moving under the influence of gravity and a background horizontal magnetic field. If we denote the local height of the fluid by $h(x, y, t)$, the horizontal velocity by $\mathbf{u}(x, y, t)$ and the horizontal magnetic field by $\mathbf{B}(x, y, t)$, the governing equations are then

$$\begin{aligned} \partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} &= -g(t) \nabla h + (\mathbf{B} \cdot \nabla) \mathbf{B}, \\ \partial_t \mathbf{B} + (\mathbf{u} \cdot \nabla) \mathbf{B} &= (\mathbf{B} \cdot \nabla) \mathbf{u}, \\ \partial_t h + \nabla \cdot (h \mathbf{u}) &= 0, \\ \vec{\nabla} \cdot (h \mathbf{B}) &= 0. \end{aligned} \quad (1)$$

Note that $g(t)$ denotes the acceleration due to a uniform in space, time-varying, vertical gravitational field acting on the fluid, and that $\mathbf{B}$ is implicitly scaled by a factor of $1/\sqrt{\mu\rho}$, where μ is the permittivity of the fluid, and ρ is the density. The resulting scaled magnetic field has units of velocity.

Stastna and Poulin [18] investigated the Faraday instability in the context of the well-known shallow-water (SW) equations. The equations they considered correspond to the above system with $\mathbf{B}$ identically zero. The authors allowed time-dependent gravity, linearized the nonlinear system and derived a single equation (their eq. 7), describing the temporal evolution of the spectral amplitude of the free surface. We shall follow the same procedure with an imposed magnetic field. Specifically, our gravity shall have the form

$$g(t) = g_0 G(t) = g_0 (1 + a \cos(\Omega t)), \quad (2)$$

with $0 < a < 1$ to prevent negative gravity.

To understand the onset of the Faraday instability in our system, we linearize about a static background state with constant magnetic field strength B_{0R} , and height H using (u, v) , (b_1, b_2) , and η as perturbations. The govern-

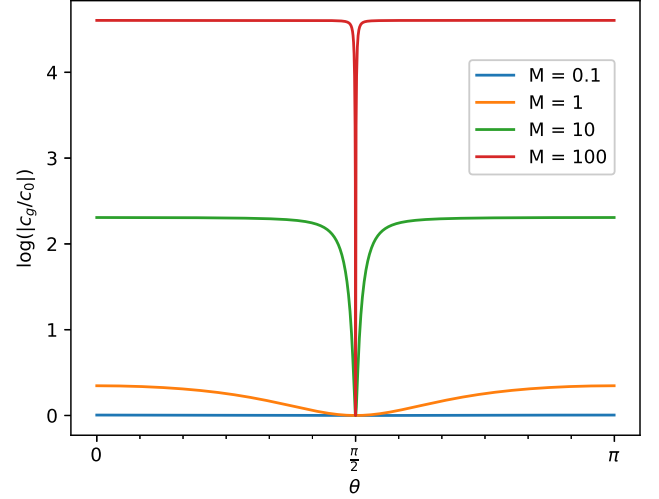


Fig. 1: The log of the magnitude of the group velocity scaled by c_0 as a function of propagation angle (θ) for different M . In all cases, the slowest direction is perpendicular to the background magnetic field (where $|\mathbf{c}_g| = c_0$, and the fastest direction is parallel to the field. As $M \rightarrow \infty$ the group speed is increasingly large and isotropic, with the exception of $\theta \approx \pi/2$.

ing equations become

$$\begin{aligned} \partial_t \eta + H \partial_x u + H \partial_y v &= 0, \\ B_{0R} \partial_x \eta + H \partial_x b_1 + H \partial_y b_2 &= 0, \\ \partial_t u + g(t) \partial_x \eta - B_{0R} \partial_x b_1 &= 0, \\ \partial_t v + g(t) \partial_y \eta - B_{0R} \partial_x b_2 &= 0, \\ \partial_t b_1 &= B_{0R} \partial_x u \\ \partial_t b_2 &= B_{0R} \partial_x v. \end{aligned}$$

These can be combined into a single wave equation for η ,

$$\partial_{tt} \eta - c_0^2 ((G(t) + M^2) \partial_{xx} \eta + G(t) \partial_{yy} \eta) = 0, \quad (3)$$

where

$$c_0 = \sqrt{g_0 H}, \quad \text{and} \quad M = B_{0R}/c_0,$$

are the gravity wave speed, and a modified Cowling/Froude Number. This equation immediately suggests that the magnetic field causes the evolution of the free surface to be anisotropic in spectral space. Indeed, if we assume a plane wave solution of the form $\eta = \hat{\eta} \exp(i(kx + ly - \omega t))$ for the constant gravity case ($a = 0$), then we get the following dispersion relation,

$$\omega^2 = c_0^2 k^2 + B_{0R}^2 k^2. \quad (4)$$

The group velocity is then

$$\mathbf{c}_g = \frac{c_0}{\sqrt{1 + M^2 \cos^2 \theta}} ((1 + M^2) \cos \theta, \sin \theta),$$

where $l/k = \tan(\theta)$. The magnitude of the group velocity is visualized in fig.1. Since θ is the angle of propagation relative to the background magnetic field, the group velocity

indicates that the energy of a disturbance is transported fastest when it travels parallel to the magnetic field, and slowest when it travels perpendicularly to the magnetic field.

As we shall see in the next two sections, when gravity is varied this anisotropy manifests as preferentially unstable directions of propagation in addition to preferentially unstable wavevectors.

Linear Stability. – In eq. (4) the RHS is positive, so, as is expected, the constant gravity system is stable for all parameters. This result does not extend to the time dependent case, because the resultant dispersion relation requires that ω is also time dependent. The system therefore does not admit plane waves as solutions. Fortunately, it is still permissible to substitute

$$\eta = \hat{\eta}(t) \exp(i(kx + ly))$$

into eq. (3), providing

$$\frac{d^2 \hat{\eta}}{dt^2} + c_0^2(k^2 + l^2) \left(1 + M^2 \frac{k^2}{k^2 + l^2} + a \cos(\Omega t) \right) \hat{\eta} = 0, \quad (5)$$

which describes the evolution of $\hat{\eta}$ at each point in spectral space. We can further simplify this equation by nondimensionalizing with a horizontal length scale L , and a time scale of $T = \frac{L}{c_0}$ to get²

$$\frac{d^2 \hat{\eta}}{ds^2} + r^2 \left(1 + M^2 \cos^2 \theta + a \cos \left(\frac{s}{Ro} \right) \right) \hat{\eta} = 0,$$

where $r^2 \equiv |\tilde{\mathbf{k}}|^2 = |L\mathbf{k}|^2$, $\cos^2 \theta = \frac{\tilde{k}^2}{r^2}$, and $s = t/T$. As a consequence of the temporal scaling we now have a Rossby number $Ro = c_0/(\Omega L)$. Finally, if we define $\delta = r^2(1 + M^2 \cos^2 \theta)$, and $\epsilon = r^2 a$ we arrive at

$$\frac{d^2 \hat{\eta}}{ds^2} + \left(\delta + \epsilon \cos \left(\frac{s}{Ro} \right) \right) \hat{\eta} = 0, \quad (6)$$

which is Mathieu's equation.

The stability of this equation with $Ro = 1$ is discussed by Kovacic, *et. al.* [4], but their approach can be generalized for other values. Specifically, they find $\delta - \epsilon$ space to be partitioned into stable and unstable regions, with unstable regions (or tongues) bounded by a pair of curves intersecting at a common point on the δ -axis. Expressions for these curves can be derived via harmonic balancing. Floquet theory implies that solutions corresponding to points on the transition curves are either $2\pi Ro$ or $4\pi Ro$ -periodic. As a result, these transition solutions can be represented by the following Fourier series:

$$\hat{\eta}_{\text{crit}}(s) = \sum_{n=0}^{\infty} a_n \cos \left(\frac{ns}{2Ro} \right) + b_n \sin \left(\frac{ns}{2Ro} \right).$$

Substituting this expression into eq. (6), and eliminating terms with common frequencies produces four infinite

determinants, each of which must be zero to avoid trivial Fourier coefficients. Approximations to the transition curves can then be found by substituting

$$\delta(\epsilon) = \delta^{(0)} + \epsilon \delta^{(1)} + \epsilon^2 \delta^{(2)} + \mathcal{O}(\epsilon^3)$$

into a finite truncation of the determinants. Doing so generates the following equations, for $n = 0, 1, 2, 3, \dots$

$$\delta_{n\pm}(\epsilon) = \left(\frac{n\gamma}{2} \right)^2 + \sum_{m=1}^{\infty} c_{n\pm, m} \gamma^{2(1-m)} \epsilon^m,$$

where the real constants $c_{n\pm, m}$ determine the left and right (corresponding to the negative and positive subscripts, respectively) boundaries of the n^{th} unstable tongue. By setting $\epsilon = 0$, we see that the leading order terms are the δ -intercepts of the transition curves. In our system $\delta = r^2(1 + M^2 \cos^2 \theta)$ and $\epsilon = r^2 a$, so after substituting and rearranging we get

$$(1 + M^2 \cos^2 \theta) r^2 - \sum_{m=1}^{\infty} c_{n\pm, m} (Ro)^{2(m-1)} (ar^2)^m = \left(\frac{n}{2Ro} \right)^2 \quad (7)$$

as the boundaries for the n^{th} unstable region. Notice that these equations no longer describe curves in an abstract $\delta - \epsilon$ space, but now describe pairs of closed curves in the spectral space of our system. At leading order these equations imply

$$\cos^2 \theta = \frac{\left(\frac{n}{2Ro} \right)^2 - 1}{M^2},$$

and each harmonic corresponds to ellipses along which $\hat{\eta}$ may go unstable. Notice that in the limit of $M^2 \rightarrow \infty$ points with $\theta = \pm \frac{\pi}{2}$ at any radius may go unstable. As a consequence, the spectrum of $\hat{\eta}$ is expected to become concentrated along the $\tilde{l}$ -axis in spectral space, which in turn implies that the structure of η in the physical domain will be dominated by features periodic in the y -direction.

To consider the stability for finite (i.e. not in the asymptotic limit) parameters, let us first consider the $\mathcal{O}(a)$ equations of the transition curves. Only the $n = 1$ region has transition curves with a non-zero first order correction; these curves are approximated by

$$\left(1 + M^2 \cos^2 \theta \pm \frac{a}{2} \right) r^2 = \left(\frac{1}{2Ro} \right)^2, \quad (8)$$

providing two concentric ellipses as the $\mathcal{O}(a)$ boundaries for this unstable region. Notice that in the limit $a \rightarrow 0$, the region converges to a single ellipse,

$$(1 + M^2 \cos^2 \theta) r^2 = \left(\frac{1}{2Ro} \right)^2,$$

which corresponds to the δ -intercept of the $n = 1$ transition curve. In the first quadrant (where $\tilde{k}, \tilde{l} \geq 0$), when

²The scale of $\hat{\eta}$ is irrelevant due to the linearity of the equation.

$n = 1$, the unstable region is bounded by the curves

$$\tilde{l}_{in}(\tilde{k}) = \sqrt{\frac{(1/2Ro)^2 - \tilde{k}^2(1 + M^2 + a/2)}{1 + a/2}} \quad (9)$$

$$\tilde{l}_{out}(\tilde{k}) = \sqrt{\frac{(1/2Ro)^2 - \tilde{k}^2(1 + M^2 - a/2)}{1 - a/2}}. \quad (10)$$

which converge to each other in the limit of $a \rightarrow 0$. These are plotted in red in fig. 2, in addition to the boundaries of the second unstable region, which were found using the second order approximation.

Evolution of the Linearized Free Surface. – In this section, we confirm the stability regions we found above analytically using numerical simulation. In particular, we'll use the linear evolution of eq. (6) for examples of different wavevectors to show that indeed the unstable regions of evolution are bounded by the ellipses given in eq. (7). Recall we had the linear equation

$$\frac{d^2 \hat{\eta}}{ds^2} + r^2 (M^2 \cos^2(\theta) + G(s)) \hat{\eta} = 0,$$

where we defined the quantities

$$\delta = r^2(1 + M^2 \cos^2(\theta)), \quad G(s) = 1 + a \cos\left(\frac{s}{Ro}\right).$$

such that we can write (where again $\epsilon = r^2 a$)

$$\frac{d^2 \hat{\eta}}{ds^2} + \left(\delta + \epsilon \cos\left(\frac{s}{Ro}\right)\right) \hat{\eta} = 0.$$

or in a more useful form, with $\eta_1 = \hat{\eta}$, $\eta_2 = \frac{d\hat{\eta}}{ds}$,

$$\frac{d}{ds} \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -(\delta + \epsilon \cos(\frac{s}{Ro})) & 0 \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix}. \quad (11)$$

This system is easily numerically integrated in time such that we can find the evolution of the time-dependent amplitude $\hat{\eta}$ for different non-dimensional wavenumbers, $\tilde{\mathbf{k}}$. However, what we wish to compute is the growth of the general solution after one period for each wavenumber. To do so, construct the fundamental solution matrix

$$F(s) \equiv \begin{bmatrix} \eta_{11}(s) & \eta_{12}(s) \\ \eta_{21}(s) & \eta_{22}(s) \end{bmatrix}, \quad (12)$$

satisfying the initial conditions that $F(0)$ is equal to the identity matrix. The Monodromy matrix is then

$$C \equiv F(T) = \begin{bmatrix} \eta_{11}(T) & \eta_{12}(T) \\ \eta_{21}(T) & \eta_{22}(T) \end{bmatrix}, \quad (13)$$

where the stability of the system is determined by the eigenvalues of C (the Floquet multipliers) satisfying

$$\lambda^2 - (\text{tr}C)\lambda + \det C = 0. \quad (14)$$

The system is said to be unstable when one of the eigenvalues has a modulus greater than one. The specific form of eq. (6) gives the useful property that $\det C = 1$ is constant such that

$$\lambda_{1,2} = \frac{1}{2} \left(\text{tr}C \pm \sqrt{\text{tr}C^2 - 4} \right). \quad (15)$$

Then, if $|\text{tr}C| < 2$ the pair of conjugate roots lie on the unit circle which yield a stable case. On the other hand when $|\text{tr}C| > 2$, one eigenvalue must have modulus smaller than 2 while the other must have modulus larger than 2 which corresponds to an unstable case [4]. We can then use the condition of $|\text{tr}C| > 2$ as a mask to determine the regions of instability and plot the largest Lyapunov exponents μ_{max} , for each wavevector $(\tilde{k}, \tilde{l})$ given by

$$\mu_{max} = \frac{\ln(\max_{\lambda}(\Re(\lambda)))}{T}. \quad (16)$$

as a measure of the growth rate in the unstable regions.

Using Heun's method, for each wavevector we evolve the fundamental solution matrix $F(s)$ using eq. (11) over a single period $T = 4\pi Ro$, and calculate the Lyapunov exponents. In the simulation, we use the physical parameters $M^2 = 1$, $a = 0.5$, $Ro = 0.5$ for simplicity which includes the influence of both gravity and magnetism. Other sets of parameters were considered and tended to similar results. Contours of the Lyapunov exponents, as given in eq. 16, are plotted in fig. 2 (in blue/green) along with the predicted transition curves (in red) from the theoretical analysis above for $n = 1, 2$. We see clear agreement between the computed growth rates μ_{max} with our derived transition curves for the $n = 1, 2$ instability tongues. The $n = 3$ unstable tongue is also present in the top-right region of the figure. We see that the $n = 1$ instability tongue is most unstable (providing the largest growth rates) while the $n = 3$ tongue gives the smallest growth for this range of wavenumbers. There are of course many other instability tongues for larger $(\tilde{k}, \tilde{l})$ but their growth rates are smaller.

More importantly, the instability tongues are wider and instabilities are faster growing along the $\tilde{l}$ axis (waves propagating perpendicularly to the magnetic field) compared to those along the $\tilde{k}$ axis (waves propagating along the magnetic field). Thus, there is a smaller range of unstable wavenumbers parallel to the magnetic field, and the growth rates within these ranges are smaller than the wave in the same instability tongue propagating in the perpendicular direction. The growth of these waves thus depends on the direction of propagation relative to the magnetic field. This provides clear evidence of anisotropy in the parametric instability where the magnetic field stabilizes the system.

To confirm this anisotropy, we also include the temporal evolution of the $\hat{\eta}$ field for the same set of parameters in fig. 3. Specifically, for the same evolution of $F(s)$ in the system (11) now for four periods, we plot $\eta_{11}(t)$ (in blue) and $\eta_{12}(t)$ (in red) for the wavenumbers $\tilde{\mathbf{k}}$ corresponding

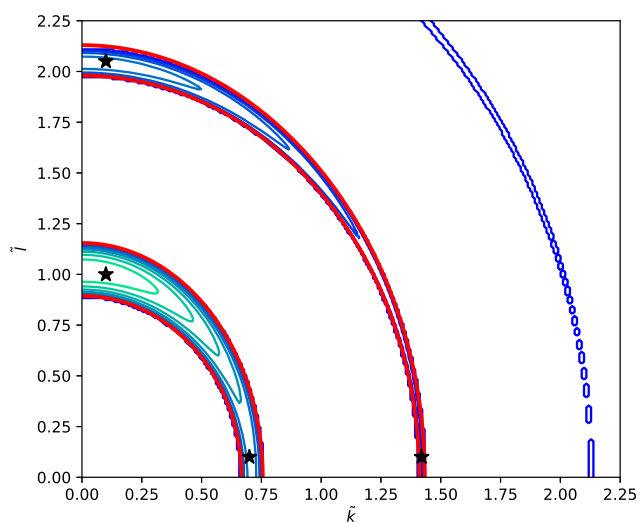


Fig. 2: Contours of the growth rates (Lyapunov exponents) for $M^2 = 1$, $a = 0.5$, $Ro = 0.5$. The computed unstable regions (blue/green contours) match the predicted transition curves (red) from the theoretical analysis above for $n = 1, 2$. The outline of the $n = 3$ unstable region is visible in the top right corner. Maximum value is 0.0444 with contour increments of 0.005.

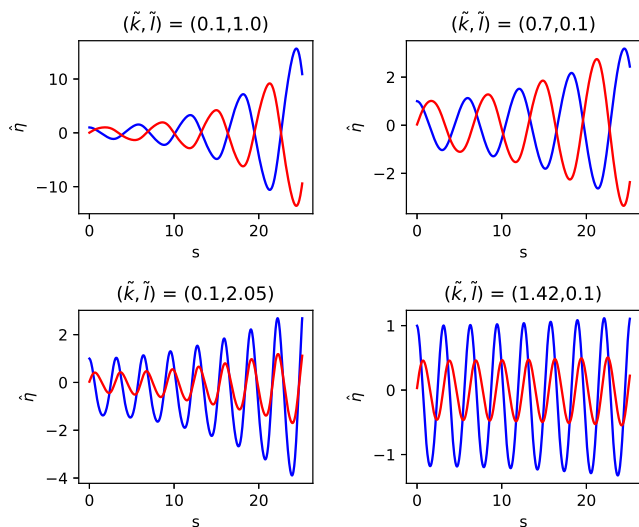


Fig. 3: Evolution of $\hat{\eta}$ for the four points depicted in fig. 2 (black stars). Top row corresponds to the first instability tongue ($n = 1$) and the bottom row to the $n = 2$ tongue. Solution was evolved for 4 periods where the blue curve corresponds to the $[1, 0]^T$ initial condition and the red curve corresponds to the $[0, 1]^T$ initial condition. In the $n = 1$ tongue, solutions are out of phase while in the $n = 2$ tongue, solutions appear in phase by the final time.

to the black stars in fig. 2. The top row of fig. 3 shows the evolution of the field in the $n = 1$ instability tongue and confirms that it is the most unstable with the largest

growth near $\tilde{\mathbf{k}} = (0, 1)$. The bottom row depicts the corresponding fields from the $n = 2$ instability tongue where the waves are still unstable but grow more slowly than the largest growing wavenumber mentioned above. Another interesting feature in the evolution features solutions that are out of phase in the $n = 1$ tongue, while showing solutions that appear in phase in the $n = 2$ tongue by the final time.

Based on the results in fig. 2, we found that waves propagating in the direction of the background magnetic field are more stable than those propagating orthogonal to it. In other words, waves parallel to the magnetic field are excited the least and those that are orthogonal are excited the most. We emphasize that this is a conservative system and there is no dissipation involved, as would occur in a system with magnetic diffusion. This type of anisotropic behaviour is also observed when magnetic diffusion is present, for example in Rayleigh-Bénard convection [25], but clearly the physical mechanism is very different. The dispersion relation (eq. 4) tells us that free waves are anisotropic due to magnetic tension, as seen in the context of the Kelvin-Helmholtz instability [26]. Hence, resonance or instability characteristics would be expected to exhibit anisotropy as well.

Conclusion. – In this work, we investigated the linear stability of the SMHD equations in vertically oscillating gravity and a constant zonal magnetic field. We found that the rest state is unstable for $a \neq 0$ and that the magnetic field introduced anisotropy in the spatial structure of the solution. In particular, in the presence of a magnetic field, the growth rates of the waves depend on their direction of propagation. Waves propagating in the same direction as the magnetic field were found to grow more slowly as compared to waves propagating in a perpendicular direction which provides evidence that the magnetic field is stabilizing. Future research directions may include the testing of a full range of parameters in both the linear and nonlinear evolution to find more compelling and more detailed evidence of the anisotropy we derived. The nonlinear regime would however require some reasonable model of energy dissipation in order to tame the instability.

MS and FJP were supported by NSERC Discovery Grants RGPIN-311844-37157 and RGPIN-2020-04084, respectively.

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